COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	15/10/26	**DATE SUBMITTED:**	15/10/26
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ACTIVITY NO. 11.1 **Containerization**

1. PROCEDURE OUTPUT

1. Create a new repository for this activity.

```
iquin@workstation:~$ cd CPE-212/
iquin@workstation:~/CPE-212$ ls
ACT_11.1                filepermissions.
```

- Instead of creating a new repository, I just add a new directory to my existing repository.

2. Install Docker and enable the docker socket.

Dependencies:

```
iquin@workstation:~$ sudo apt install containerd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and a
```

```
.quin@workstation:~$ sudo apt install -y docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
iquin@workstation:~$ sudo systemctl enable --now docker
Synchronizing state of docker.service with SysV service script with /usr/li
temd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
iquin@workstation:~$ sudo systemctl enable --now containerd
iquin@workstation:~$
```

```
iquin@workstation:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset:
     Active: active (running) since Thu 2026-10-15 02:08:24 UTC; 5min ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 3677 (dockerd)
      Tasks: 9
     Memory: 33.8M (peak: 35.7M)
        CPU: 699ms
     CGroup: /system.slice/docker.service
             └─3677 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/co

Oct 15 02:08:23 workstation dockerd[3677]: time="2026-10-15T02:08:23.65528410
```

```
iquin@workstation:~$ sudo systemctl status containerd
● containerd.service - containerd container runtime
     Loaded: loaded (/usr/lib/systemd/system/containerd.service; enabled; pr
     Active: active (running) since Thu 2026-10-15 02:06:03 UTC; 7min ago
       Docs: https://containerd.io
   Main PID: 3477 (containerd)
      Tasks: 7
     Memory: 21.5M (peak: 22.3M)
        CPU: 785ms
     CGroup: /system.slice/containerd.service
             └─3477 /usr/bin/containerd
```

```
iquin@workstation:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3~24.04.2
iquin@workstation:~$
```

```
iquin@workstation:~$ sudo docker info
Client:
 Version:    29.1.3
 Context:    default
 Debug Mode: false
 Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.38.0
    Path:     /usr/libexec/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version:  v5.6.0
    Path:     /usr/libexec/docker/cli-plugins/docker-compose
  trust: Manage trust on Docker images (Docker Inc.)
    Version:  29.1.3
    Path:     /usr/libexec/docker/cli-plugins/docker-trust

Server:
 Containers: 0
  Running: 0
  Paused: 0
  Stopped: 0
 Images: 1
```

```
iquin@workstation:~$ ls -l /var/run/docker.sock
srw-rw---- 1 root docker 0 Oct 15 02:08 /var/run/docker.soc
iquin@workstation:~$
```

Analysis: I encountered an error, I need to manually install containerd before installing docker.io

3. Add a Docker group to your current user.

```
quin@workstation:~$ sudo usermod -aG docker "$USER"
quin@workstation:~$
```

```
quin@workstation:~/CPE-212/ACT_11.1$ newgrp docker
```

```
quin@workstation:~/CPE-212/ACT_11.1$ docker run hello-world
nable to find image 'hello-world:latest' locally
atest: Pulling from library/hello-world
f55086f7dd0: Pull complete
5e71e642bf5: Download complete
igest: sha256:11a3eb1597e864c51b3f56733cb2e8a90bf0c284bfc05f9444d81c558bc8
tatus: Downloaded newer image for hello-world:latest

ello from Docker!
his message shows that your installation appears to be working correctly.

o generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs th
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent
    to your terminal.

o try something more ambitious, you can run an Ubuntu container with:
```

Analysis: usermod is the easiest way to add docker group to my system.

4. Create a Dockerfile to install web and DB server.

```yaml
---
- name: Build the web and database container image
  hosts: localhost
  become: true

  tasks:
    - name: Install Docker and Python Docker support
      ansible.builtin.apt:
        name:
          - docker.io
          - python3-docker
        state: present
        update_cache: true

    - name: Enable and start Docker
      ansible.builtin.service:
        name: docker
        state: started
        enabled: true
```

```yaml
- name: Build the Docker image
  community.docker.docker_image:
    name: activity11-web-db
    tag: "1.0"
    source: build
    build:
      path: "{{ playbook_dir }}"
```

Analysis: Installing docker for Web and DB server using ansible. This will automate the system to install both docker.

5. Install and build the Dockerfile using Ansible.

```
---
- name: Build the web and database container image
  hosts: localhost
  become: true

  tasks:
    - name: Install Docker and Python Docker support
      ansible.builtin.apt:
        name:
          - docker.io
          - python3-docker
        state: present
        update_cache: true

    - name: Enable and start Docker
      ansible.builtin.service:
        name: docker
        state: started
        enabled: true
```

```
PLAY RECAP *********************************************************************
localhost                  : ok=4    changed=1    unreachable=0    failed=0
kipped=0    rescued=0    ignored=0
```

6. Add, commit and push it to your repository

```
iquin@workstation:~/CPE-212/ACT_11.1$ git add .
iquin@workstation:~/CPE-212/ACT_11.1$ git commit -m "HOA 11"
[main 4d3b938] HOA 11
 1 file changed, 17 insertions(+), 71 deletions(-)
iquin@workstation:~/CPE-212/ACT_11.1$ git push origin main
Username for 'https://github.com': ErickAlvarezz
Password for 'https://ErickAlvarezz@github.com':
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 675 bytes | 675.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/ErickAlvarezz/CPE-212
   25451b3..4d3b938  main -> main
iquin@workstation:~/CPE-212/ACT_11.1$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What are the benefits of implementing containerizations?
    - Containerization helps make applications easier to install, test, and run. It packages an application and its required software into a container image, which helps reduce problems caused by differences between environments.

Conclusion: In this activity, I learned how to create a Dockerfile, install and configure Docker, and use Ansible to automate the building of a container image. I also learned how to use Git to save and upload project files to GitHub.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Error Explanation and Grammar Correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI was only used for error explanation and grammar correction.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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